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Temperature Dependence of the Liquid Tin Resistivity

By

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The temperature dependence law of the liquid metal resistivity is usually attributed to that of the atomic structure. In the framework of the Percus-Yevick approximation applied to hard spheres, various laws of temperature dependence of their diameter are used. They are deduced from different physical or thermodynamical properties (self-diffusion, structure, entropy, isothermal compressibility) or by minimizing the free energy. A comparison with experimental measurements on liquid tin up to 1300 °C leads to a fairly good agreement.

La dépendance en température de la résistivité des métaux liquides est d'habitude attribuée à celle de la structure atomique. Dans le cadre de l'approximation de Percus-Yevick appliquée aux sphères dures, différentes lois de dépendance en température de leur diamètre ont été employées. Elles ont été obtenues à partir de diverses propriétés physiques ou thermodynamiques (autodiffusion, structure, entropie, compressibilité isotherme) ou par minimalisation de l'énergie libre. La comparaison avec nos mesures sur l'étain liquide jusqu'à 1300 °C est discutée et révèle un accord globalement satisfaisant.

1. Introduction

Information about electrical resistivity of liquid metals has been made available for many years and there has been a number of attempts to explain the experimental data. It is unquestionable that the nearly-free-electron model is one of the most useful for describing the electrical resistivity in simple liquid metals. In this paper we have computed the resistivity of liquid tin according to Ziman's formula [1], and compared it with our experimental measurements at different temperatures. If the use of Ziman's formula is successful in explaining the measured resistivities of liquid metals and their temperature dependence, it must be noted that this formula is very sensitive to the ingredients, approximations and assumptions, namely

(i) the pseudopotential concept to describe the electron–ion interaction. In the present work the pseudopotential has been chosen to be of the local and energy-independent Ashcroft [2] form;

(ii) the dielectric function in order to screen the bare-ion potential. We use the simple and convenient analytical form proposed by Vashishta and Singwi [3];

(iii) the structure factor which depends on the packing fraction of the liquid metal, if we make use of the hard-sphere (HS) model to represent the ionic structure [4].

It is well known that the HS model may be employed in the study of both thermodynamic properties [5] and atomic transport processes [6, 7]. Although the hard-sphere model is probably not completely satisfactory to describe the structure factor of liquid tin because of the shoulder on the right-hand side of the main peak, this model can be used to take into account its temperature dependence in a simple manner.

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The basic steps of the Ziman theory are outlined in Section 2. The temperature dependence of the packing fraction is discussed in Section 3, with different temperature dependence laws for the hard-sphere diameter. Finally the comparison of the calculated resistivities of liquid tin with our experimental results, which leads to a fairly good agreement, is displayed in Section 4.

2. The Electrical Resistivity of Liquid Metals

For simple metals the nearly-free-electron (NFE) model appears to be reasonably well based and the well-known Born approximation formula proposed by Ziman [1] is given by

$$\varrho = \frac{3}{4}\frac{\pi m^2 \Omega}{\hbar^3 e^2 k_F^6}\int_0^{2k_F} a(q)\, w(q)^2\, q^3\, dq\,, \tag{1}$$

where Ω is the atomic volume, $a(q)$ the structure factor, $w(q)$ the model potential form factor, and k_F the value of the Fermi wave vector defined by the equation

$$k_F^3 = \frac{3\pi^2 z}{\Omega}, \tag{2}$$

where m, e, $\hbar$, π are the usual constants and z is the valence. In order to compute the resistivity, we must choose an appropriate model potential and structure factor.

2.1 The model potential

We employ the Ashcroft [2] simple local empty core potential (ECP)

$$w(q) = \frac{-ze^2}{q^2 \Omega \varepsilon(q)} \cos(qR_c)\,, \tag{3}$$

where $\varepsilon(q)$ is the dielectric screening function, R_c the empty core radius. Rather than fitting, more or less arbitrarily, R_c on a macroscopic experimental property (resistivity, thermoelectric power of liquid metal, or phonon spectra) we choose R_c so as to have the same node as the Animalu-Heine (AH) [8] non-local model potential ($R_c = 0.589 \times 10^{-10}$ m). This is preferable, because the model parameters occurring in the AH potential are determined independently by matching the wave functions. At the same time the ECP is easier to handle.

Exchange and correlation corrections for the free electrons are incorporated into the reliable form of the screening function $\varepsilon(q)$ through the local-field correction $G(q)$,

$$\varepsilon(q) = 1 + [1 - G(q)]\, k_s^2 \frac{f(q)}{q^2}, \tag{4}$$

where

$$k_s^2 = \frac{4k_F m^*}{\pi a_0 m}. \tag{5}$$

a_0 is the Bohr radius, m^*/m the effective mass ratio taken to be unity, and $f(q)$ is the Lindhard function

$$f(q) = \frac{1}{2} + \frac{1 - (q/2k_F)^2}{4(q/2k_F)} \ln\left|\frac{1 + q/2k_F}{1 - q/2k_F}\right|. \tag{6}$$

The function $G(q)$ accounting for exchange and correlation amongst the conduction electrons is given by Vashishta and Singwi (VS) according to

$$G(q) = A\left[1 - \exp\left(\frac{-Bq^2}{k_F^2}\right)\right], \tag{7}$$

where A and B are functions of r_s which is the radius of a sphere whose volume is equal to the average volume per electron,

$$r_s = \left(\frac{\Omega}{z}\frac{3}{4\pi}\right)^{1/3}. \tag{8}$$

It is known that the resistivity is sensitive to the prescription used for including exchange and correlation contributions. However, the analytical form suggested by VS represents an accurate fit to their self-consistent screening function and the parameters A and B tabulated in their paper have a weak dependence on the electron density.

2.2 The structure factor

The temperature dependence of the structure factor can be drawn from the experiments, but it can also be computed according to the Percus and Yevick [9] theory applied to the hard-sphere fluid. The HS model structure factor depends strongly on the packing fraction η of the liquid metal, which is given by

$$\eta(T) = \frac{\pi\sigma^3(T)}{6\Omega(T)}, \tag{9}$$

where $\sigma(T)$ is the HS diameter as a function of temperature and $\Omega(T)$ the atomic volume deduced from the experimental density compiled by Crawley [10].

3. Different Choices for the HS Diameter

Several temperature dependence laws for the hard-sphere diameter can be chosen.

a) The HS diameter is fixed. Very often the HS diameter is held constant in electronic property calculations. We have used the value ($\sigma = 2.854 \times 10^{-10}$m) corresponding to the packing fraction $\eta = 0.43$ at 250 °C, which has been deduced from X-ray diffraction experiments by Waseda [11].

b) The HS diameter is varied according to the self-diffusion measurements carried out by Bruson and Gerl [6],

$$\sigma(T) = 3.16 \times 10^{-10}\left[1 - 0.082\left(\frac{T}{T_M}\right)^{1/2}\right], \tag{10}$$

where T_M is the melting point temperature and T the temperature of the liquid (in K). Alternatively, when no experimental data are available, we suggest using the formula of Protopapas and Parlee [7],

$$\sigma(T) = 3.3 \times 10^{-10}\left[1 - 0.112\left(\frac{T}{T_M}\right)^{1/2}\right]. \tag{11}$$

c) The packing fraction, hence σ, is deduced from the entropy [12] computed at different temperatures. The experimental entropy is calculated as follows:

$$S_T = S_{298K} + \int_{298K}^{T_M} \frac{C_p^{sol}}{T}\,dT + \frac{L_M}{T_M} + \int_{T_M}^{T} \frac{C_p^{liq}}{T}\,dT. \tag{12}$$

S_{298K}, $C_p^{sol}(T)$, $C_p^{liq}(T)$, L_M are taken from Kubaschewski et al. [13]. The entropy is related to η by the equation based on the Carnahan and Starling (CS) [14] relation

$$\frac{S_T}{Nk_B} = \ln(\lambda^3\Omega) + \frac{5}{2} + \frac{3\eta^2 - 4\eta}{(1-\eta)^2} \tag{13}$$

with

$$\lambda = \left(\frac{Mk_BT}{N2\pi\hbar^2}\right)^{1/2}. \tag{14}$$

The de Broglie thermal wavelength λ which is connected to the atomic mass M as well as to various universal constants, justifies the classical treatment of static properties of the liquid metals.

By solving (12) and (13) together we obtain the variation of σ versus T.

In this work, we have calculated σ with the Carnahan and Starling equation of state which is very close to the one derived from the molecular dynamics. Ohkoshi et al. [15] have also discussed this method in connection with the Percus-Yevick phonon theory. However, the electronic term can be neglected because it is very small, as could be seen on their Table 1.

d) The packing fraction, hence σ, is obtained from the experimental isothermal compressibility which is deduced by using thermodynamic relations

$$\chi_T = \chi_A + \frac{TV\alpha^2}{C_p} \tag{15}$$

and

$$\chi_A = \frac{1}{du^2}; \tag{16}$$

χ_A is the adiabatic compressibility, V the molar volume, C_p the specific heat taken from Kubaschewski et al. [13], and u the velocity of sound taken from Webber and Stephens [16]. The density d and the thermal expansion coefficient α are deduced from Crawley [10].

If the HS fluid model is used with the CS equation of state, η is related to the compressibility [5] by the relation

$$\chi_T = \frac{(1-\eta)^4}{2\eta(4-\eta) + (1-\eta)^4} \frac{\Omega}{k_BT}. \tag{17}$$

For a real liquid metal, however, we have to take into account the kinetic energy E_F of the electron gas and the electron–ion interaction. This was done in the same way as Ascarelli [17],

$$\chi_T = \left[\frac{k_BT}{\Omega D(\eta)} + \frac{2}{3}\frac{zE_F}{\Omega} - \frac{4zB(\eta)}{9\Omega^{4/3}}\right]^{-1}, \tag{18}$$

where

$$D(\eta) = \frac{(1-\eta)^4}{2\eta(4-\eta) + (1-\eta)^4}. \tag{19}$$

The term $B(\eta)$ has been obtained by Ascarelli at the melting point and was held constant. We have in fact let it vary with temperature by writing it as a function of η,

$$B(\eta) = \frac{3}{z}\Omega^{1/3}k_BTC(\eta) + \frac{6}{5}\Omega^{1/3}E_F \tag{20}$$

with

$$C(\eta) = \frac{1 + \eta + \eta^2 - \eta^3}{(1 - \eta)^3}. \tag{21}$$

At any temperature other than the melting point, the packing fraction η, and of course σ, can be determined numerically with the aid of (15) and (18).

e) The HS diameter is obtained by minimizing the free energy. The HS diameter of liquid tin is calculated by using a pseudopotential model together with an application of thermodynamic perturbation theory based on the variational method. According to the approach of Mansoori and Canfield [18], the Helmholtz free energy of a metal can be written in the form of the Gibbs-Bogolyubov inequality

$$F < F_H + \langle E - E_H \rangle_H , \tag{22}$$

where F_H is the free energy of the hard sphere (H) reference system with potential energy E_H, and $\langle E - E_H \rangle_H$ denotes the expected mean value weighted by exp $(-E_H/k_B T)$. If the reference system is a hard-sphere fluid, the diameter of the spheres may be chosen to yield a minimum Helmholtz free energy. In the evaluation of the free energy [19], the electron–ion interaction has been treated within the framework of the second-order perturbation theory through the ECP model combined with the simple form of the VS dielectric function. The entropy term has been given by the CS equation of state.

f) The HS diameter is obtained from the experimental structure factor. The packing fraction η has been fitted by Waseda [20] on the X-ray diffraction data and the temperature dependence is approximated by the exponential function

$$\eta(T) = A \exp(-BT) . \tag{23}$$

The coefficients A and B for liquid tin are 0.488 and 2.57×10^{-4} K^{-1}, respectively.

4. Experimental Results and Discussion

4.1 Measurement of resistivity

The electrical resistivity of tin was measured from the melting point to 1300 °C in a four-probe quartz cell fitted with tungsten electrodes which was calibrated with triply distilled mercury. The resistivity is measured with a precision of 0.3% and the temperatures are accurate to ± 3 K at 1000 °C. Our results are in good agreement with those of Pokorny and Aström [21] who measured carefully the experimental temperature dependence up to 700 °C, as well as with those of Davies and Leach [22] and of Mera et al. [23] up to 530 °C. The experimental method has been fully described by Gasser [24].

The experiments are reported in an analytical form, valid up to 1300 °C, by calculating the regression of the resistivity on the temperature, in °C. The coefficient of correlation is 0.9998 and the relationship is

$$\varrho = 25.758 \times 10^{-3} t + 41.825 \ \mu\Omega\text{cm} .$$

4.2 Hard-sphere diameter versus temperature

The results on the variation of the HS diameter σ versus temperature are presented in Fig. 1. As can be seen two sets of slopes are observed. The first one includes all thermodynamic results, i.e. obtained from compressibility (curve d), entropy (curve c), and self-diffusion (curve b). In the second one we find those derived from constant value of σ (curve a), and from minimizing the free energy (curve e).

The curve deduced from the compressibility stops at 500 °C, because no experiment is available beyond 500 °C and the values for σ are rather too high. Even the correction proposed by Ascarelli [17] is not sufficient to bring down the HS diameter of tin

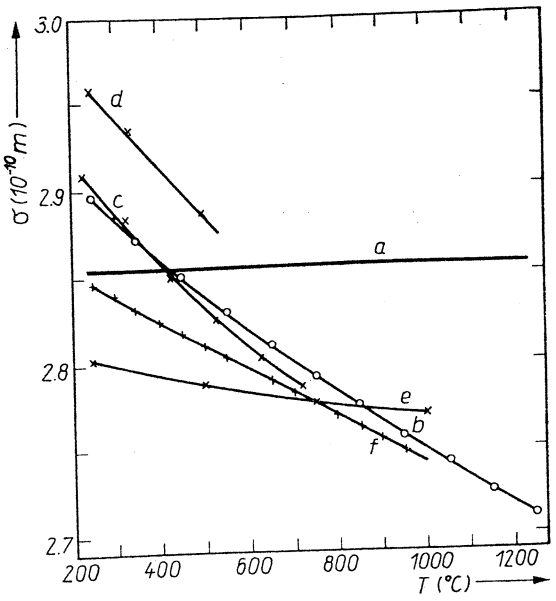


Fig. 1. HS diameter of liquid tin vs. temperature. (a) with σ constant, (b) from the self-diffusion experiments [5], (c) from the entropy measurements [13], (d) from the compressibility measurements [18], (e) by minimizing the free energy calculated with the ECP model and the VS dielectric function, (f) by fitting the experimental structure factor [20]

to a reasonable value, say from 2.95×10^{-10} to 2.85×10^{-10} m at about 250 °C. Although it already corrects the scheme excluding the free-electron-gas contribution not displayed in the figure, it provides a curve nearly parallel and close to those obtained from other thermodynamic data.

The values of σ determined by the variational calculation of F with respect to σ are also shown in Fig. 1. It could seem that there are no considerable differences in the principle of this method and the previous calculations related to the compressibility, because the latter could be regarded as an expression of the lowest order for the free energy ensuring the condition $p = (\partial F/\partial V)_T = 0$. In fact it appears that differences in details of calculations drastically affect σ, since σ as well as its slope are lower than the former results connected with the experimental thermodynamic quantities.

Finally, curve f deduced directly from the structure has an intermediate slope. It is worth recalling that the qualitative features of the curve of the tin structure factor $a(q)$ are more or less well reproduced by the HS model, though we can take advantage of that model to explain the temperature dependence of $a(q)$ in terms of $\sigma(T)$. Generally the first peak of $a(q)$ tends to become flatter and broader with increasing temperature, but its position remains nearly unchanged. For that reason Waseda [20] fitted the packing fraction $\eta(T)$ to the peak height according to (23).

4.3 Resistivity versus temperature

The resistivity of liquid tin has been computed at different temperatures, but our study begins with an investigation of the effects of the HS model structure factors, whose $\sigma(T)$ proceed from various approaches, on the resistivity. The curves are presented in Fig. 2. As for the HS diameters (Fig. 1), the results with experimental compressibility given by curve d are rather bad, though the slope of $\varrho(T)$ is certainly the best. The results obtained with experimental entropy coincide with those of the self-diffusion (curve b), but the corresponding curve is not displayed in Fig. 2. If we consider curves a and f we can see the effect of the temperature dependence of the HS model structure factor, which must be imperiously taken into account for calculating resistivity. Comparing curves e and f it is somewhat astonishing that two theories with so different starting points, i.e. minimization of the free energy and fitting on the experimental structure factor, come to such similar results concerning the resistivity.

We now turn to examine the resistivity calculated with the measured structure factor, without consideration of the model for $a(q)$, and the question which arises is whether the HS model is responsible for the observed discrepancies in the resistivity. Firstly, we ascertain that the two selected experiments of neutron and X ray diffraction for $a(q)$, labelled N (North et al. [25]) and W (Waseda [20]), can provide rather different resistivities at high temperature. Secondly, it is interesting to note the reduction of the gap bet-

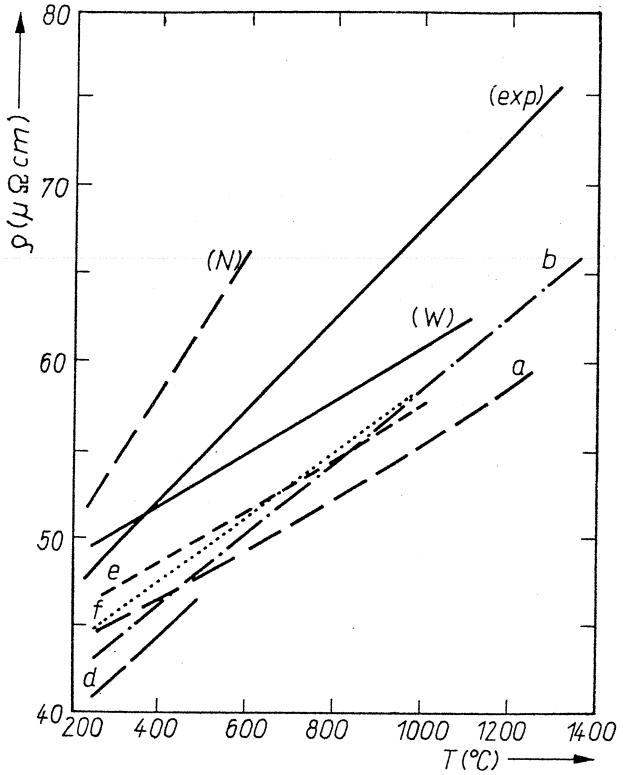


Fig. 2. The resitistivy of liquid tin vs. temperature. The curves N and W display resistivity calculated with the experimental structure factor of North et al. [25] and of Waseda [20], respectively. The other curves display resistivity obtained with the HS model structure factor, whose HS diameters are referenced as in Fig. 1

ween the resistivity issued from the $a(q)$ measured by Waseda and from the $a(q)$ fitted on it, if the temperature increases. This indicates that the arrangement of the ions in liquid tin can be modelled by the HS model at high temperature with a good degree of confidence.

Indeed nearly all the calculated values are systematically lower than the experimental one. Such discrepancies occur because the resistivity is notoriously sensitive to the choice of pseudopotential and to the form of the electron-gas response function. Here the R_c parameter in the ECP model has been chosen as previously described, and it has been kept constant through all the calculations.

5. Conclusion

An overall agreement can be found on the resistivity slopes deduced from thermodynamic data. However, results on $\sigma(T)$ extracted from compressibility are probably less accurate than those obtained from self-diffusion or entropy. The lack of knowledge of C_p and u at high temperature makes that Bruson's formula is the only practical method for computing high-temperature HS diameters. This work shows that keeping σ constant largely underestimates the resistivity. Likewise, minimizing the free energy gives also too small a temperature dependence on σ and on the resistivity temperature coefficient. However, a variation of R_c with temperature could also contribute to change the resistivity-versus-temperature curve as has been emphasized by Bellissent et al. [26] who have deduced $R_c(T)$ for gallium, by using their experimental structure factor and the experimental resistivity of Pokorny and Aström [21].

The situation at the melting temperature is already well defined with the use of a full non-local pseudopotential and of the optimized random-phase approximation for $a(q)$ [27]. Nevertheless, in order to improve the agreement between experiments and theory for resistivity at high temperature, we must dispose of very careful measurement of the tin structure factor and we also have to use a better description of the electron–ion interaction, e.g., through the ECP model by taking into account the energy dependence of the R_c parameter.

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